Effectiveness for Machine Translation Method Using Inductive Learning on Number Representation

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Abstract. On our proposed method, source language is translated into target language via Number Representation. A text in the source language is translated into a number representation text. The number representation text is the number string corresponding to the original source language text. The number representation text is translated into a number representation text for the target language. The number representation text is translated into a text in the target language. The text is the translation result finally. A number representation text is more abstract than the original text because the number representation text corresponds to several texts. The system based on our proposed method is able to acquire more translation rules on number representation than that on the original text by Inductive Learning. Moreover, the system disambiguates number representation by its own adaptability. In the experiment, the correct translation rate for our proposed method is higher than that for the method without number representation. Thus, it is proved that our proposed method is more effective for machine translation.

1 Introduction

Opportunities and needs are increasing to communicate with people in other countries because of the rapid expansion of the computer networks. However, the language problem disturbs smooth communication. The machine translation system is very effective to solve this problem.

The main method of machine translation systems is the Rule-based method. A system based on the Rule-based method is able to increase its correct translation rate when a large dictionary is given for the translation. However, it is impossible to implement the large dictionary dealing with complete linguistic phenomena to the system. Therefore, its correct translation rate is still low and the quality is poor. The Example-based and the Statistical machine translation[1][2] have been proposed to resolve this problem. A system based on these methods

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is able to increase its correct translation rate as the learning data increases. The system needs large data for the high quality translation. It is difficult that the system enables translation for various languages because there are languages without adequate corpora.

K. Araki and H. Echizen-ya et al. have proposed the machine translation method using Inductive Learning with genetic algorithms[3][4]. The system based on this method is able to learn the translation rules from the only appearance of written characters on data by Inductive Learning and generate new translation examples through crossover and mutation in genetic algorithms automatically. However, many erroneous rules have been acquired in spite of selection ability of genetic algorithms.

Our proposed method uses number representation for Inductive Learning. The system based on our proposed method acquires common and different parts as the translation rules from the inputted text by Inductive Learning on number representation. The common part is the same characters between two sentences. The different part is the sentence without the common part. The system is not able to acquire the rules from the text which does not have the common parts because the system uses the only appearance of written characters. A number representation text is more abstract than the original text because the number representation text corresponds to several texts. There are more common parts on number representation than that on the original text. Therefore, the system based on our proposed method is able to acquire more translation rules on number representation than that on the original text. Some translation rules include different texts with same meaning. These rules are effective for the translation. The system needs number representation in order to acquire these translation rules. Moreover, we have proved that the number representation text is able to be translated into the correct text[5][6]. Then, the system is able to disambiguate the number representation text by its own adaptability. Thus, the system based on our proposed method is able to learn more rules which are effective for the translation by number representation and translate into the correct text.

This paper describes our proposed method "Machine Translation Method Using Inductive Learning on Number Representation" and the results of the evaluation experiment for English-Japanese translation in our proposed method.

2 Outline of Our Proposed Method

First of all, a user inputs a source language text into the system based on our proposed method. The text is translated into a number representation text. The number representation text is the number string assigned to the original text. The number representation text for the source language is translated into a number representation text for the target language. Last of all, the number representation text for the target language is translated into a text in the target language. Thus, a source language text is translated into a target language text via number representation on our proposed method. A number representation text is more abstract than the original text because the number representa-

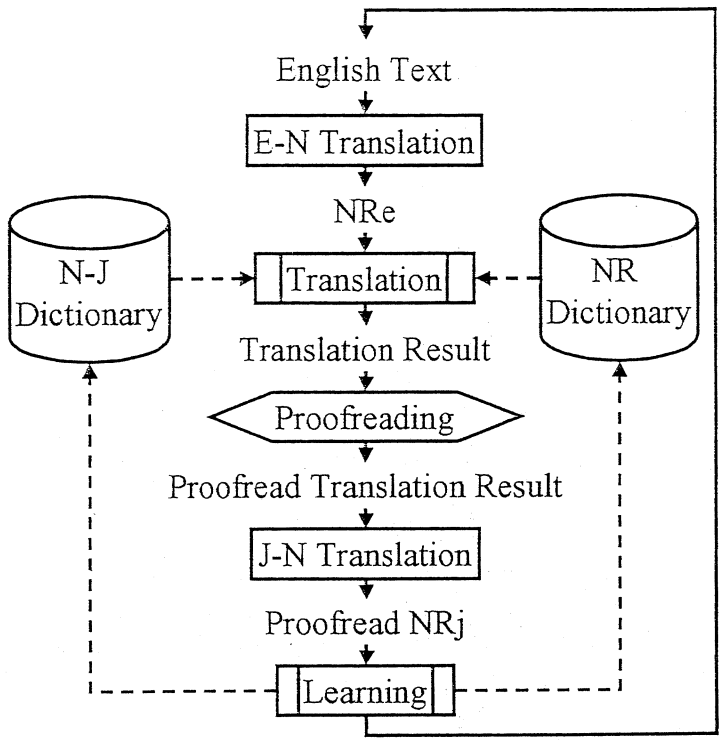


Fig. 1. Procedure

tion text corresponds to several texts. Then, the system based on our proposed method is able to acquire more translation rules on number representation than that on the original text by Inductive Learning. A translation rule is able to include different texts with same meaning when the different texts correspond to the same number representation text. The system selects the translation rules which are fit for the inputted data and translates into the target language text. Thus, the system based on our proposed method is able to efficiently learn and translate by the abstraction of the number representation.

3 Processes

Figure 1 shows the procedure for our proposed method applied to English-Japanese translation. The procedure consists of E-N Translation, Translation, Proofreading, J-N Translation and Learning Process in this order. The E-N translation process is for the translation of English into NRe. The NRe expresses number representation for English. The J-N translation process is for the translation of proofread translation result into Proofread NRj. The NRj expresses number representation for Japanese.

Table 1. Correspondence of Number to Alphabet

1:.?-'	2:ABC	3:DEF
4:GHI	5:JKL	6:MNO
7:PQRS	8:TUV	9:WXYZ
*:	0:	#:space

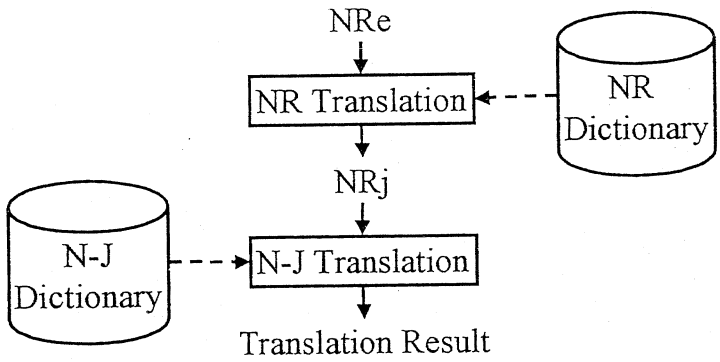


Fig. 2. Translation Process

3.1 E-N Translation Process

An English text inputted by a user is translated into the NRe in this process. Table 1 shows how to translate English characters into the number characters. For example, any of the characters "p", "q", "r" and "s" are translated into "7". The translation is able to apply to any language because this is the normal assignment for mobile phones. When the user inputs "I enjoy baseball.", this English text is translated into "4#36569#227322551".

An NRe is able to correspond to some texts and is abstract. For example, "home" is translated into "4663" and the "4663" is able to also correspond to "good" and so on. However, the system based on our proposed method disambiguates the number representation by its own adaptability. Thus, The system is able to efficiently learn and translate by the abstraction of number representation.

3.2 Translation Process

The NRe is translated into a Japanese text in this process. This process consists of NR Translation and N-J Translation Process. Figure 2 shows it. The NR translation process is for the translation of NRe into NRj. The N-J translation process is for the translation of NRj into Japanese.

NR Translation Process. The NRe is translated into an NRj by the number representation dictionary in this process. The dictionary is expressed as NR

Table 2. Translation Example

English Text	I enjoy baseball.
E-N Translation Process	
NRe	4#36569#227322551
NR Dictionary	
(English : Japanese)	
4#36569# @a 1 : @b 0324*##	
2273225 : 8281	
NR Translation Process	
NRj	82810324*##
N-J Dictionary	
8281:*yakiyuu* 0:*wo* 3:*su*	
24*:*kida* ##:.	
N-J Translation Process	
Translation Result	*yakiyuuwosukida.*
Proofreading Process	
Proofread Translation Result	*yakiyuuwotanosimu.*
J-N Translation Process	
Proofread NRj	828104537##

Dictionary in Figure 2. The NR dictionary has rules to translate NRe into NRj. The rules in the NR dictionary are acquired in the learning process. When there are some candidates for the translation, their credibility is evaluated by the credibility evaluation function(CEF). The function is defined as :

$$CEF = AF + \alpha \times CF - \beta \times EF \tag{1}$$

where α and β are coefficients. AF is the appearance frequency in text. CF is the correct translation frequency. EF is the erroneous translation frequency. CF and EF are based on the NR dictionary and are updated in the NR feedback process. The credibility for the translation is higher when AF is higher, CF is higher and EF is lower.

Table 2 shows an example of the translation. In Table 2, the NR dictionary has two rules "4#36569# @a 1 : @b 0324*##" and "22732255 : 8281". "@x" in the rule expresses a variable and is able to be replaced with another rule. Then, "@a" and "@b" are replaced with "2273225" and "8281" in Table 2. Therefore, the NRe is translated into "82810324*##".

N-J Translation Process. The NRj is translated by N-J Dictionary in this process. The N-J dictionary has rules to translate NRj into Japanese. When there are some candidates for the translation, their credibility is evaluated by the function which is similar to the equation (1). We show the detail of the

Table 3. Correspondence of Number to *Kana*

1:*a,i,u,e,o*	2:*ka,ki,ku,ke,ko*	3:*sa,si,su,se,so*
4:*ta,ti,tu,te,to*	5:*na,ni,nu,ne,no*	6:*ha,hi,hu,he,ho*
7:*ma,mi,mu,me,mo*	8:*ya,yu,yo*	9:*ra,ri,ru,re,ro*
*:Voiced Sound, P-Sound	0:*wa,wo,n*	#:Punctuation Marks

function in [6]. The ideal translation result is the Japanese text intended by the user.

Table 2 shows an example of the N-J translation. In Table 2, the N-J dictionary has five rules "8281:*yakiyuu*"[1], "0:*wo*", "3:*su*", "24*:*kida*" and "## :.". Then, the NRj "82810324*##" is translated into "*yakiyuuwosukida.*".

3.3 Proofreading Process

When the translation result has errors, the user proofreads the only appearance of written characters for them. Therefore, the proofread translation result is the Japanese text intended by the user.

In Table 2, the translation result has errors. The user replaces "*sukida*" with "*tanosimu*" to proofread the errors. The proofread translation result "*yakiyuuwotanosimu.*" is the sentence intended by the user.

3.4 J-N Translation Process

The proofread translation result is translated into the proofread NRj in this process. The proofread NRj is used in the learning process. Table 3 shows how to translate Japanese *Kana* characters into the number characters. For example, any of the *Kana* characters "*ma*", "*mi*", "*mu*", "*me*" and "*mo*" are translated into "7". This is the normal assignment for Japanese mobile phones. When the proofread translation result is "*yakiyuuwotanosimu.*", this Japanese text is translated into "828104537##". The proofread NRj is as abstract as the NRe.

3.5 Learning Process

The translation rules are acquired and registered into the NR dictionary and the N-J dictionary in this process. This process consists of NR Feedback, Re-N-J Translation and N-J Feedback Process. Figure 3 shows it. In the NR feedback process, the NR dictionary is updated. In the Re-N-J translation process, proofread NRj is again translated into Japanese. In the N-J feedback process, the N-J dictionary is updated.

[1] Japanese characters are written in *italics*. "*yakiyuu*" is generally expressed as "*yakyuu*" in Japanese. However, "*kyu*" is translated into "28" and the "28" also corresponds to "kiyu" in the system. Therefore, "*yakyuu*" is expressed as "*yakiyuu*" in this paper.

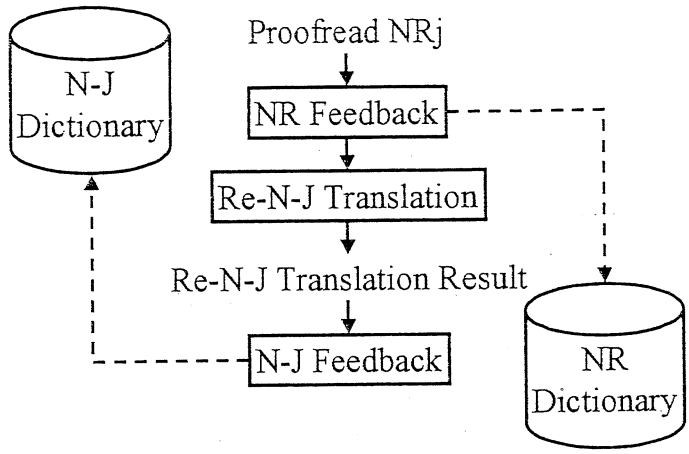


Fig. 3. Learning Process

NR Feedback Process. The NR dictionary is updated based on the NRe, the NRj and the proofread NRj in this process.

Learning for NR Dictionary. The translation rules are acquired from the NRe and the proofread NRj, and registered into the NR dictionary. The pair "NRe:Proofread NRj" is compared to the translation rules in the NR dictionary. When there are common parts between the pair and the translation rule in the NR dictionary, the common and different parts are registered as new translation rules into the NR dictionary.

Table 4 shows an example of the learning. The pair "NRe:Proofread NRj" and the translation rule in the NR dictionary are as follows:

"4#36569#227322551:828104537##"
"4#36569#8366471:45304537##"

The common parts between the pair and the translation rule are underlined. The acquired NR translation rules are shown in Table 4. "@x" in the rule expresses a variable and is able to be replaced with another rule in the translation process.

Updating for NR Dictionary. CF and EF for the translation rules are updated based on the NRj and the proofread NRj. If the proofread NRj includes a translation rule of the NRj, the translation rule is correct. Then, the system adds 1 to the value of CF for the translation rule. If the proofread NRj does not include a translation rule of the NRj, the translation rule is erroneous. Then, the system adds 1 to the value of EF for the translation rule. Thus, credibility of the translation rules is updated in the NR dictionary.

In Table 4, the proofread NRj includes "8281" of the NRj and does not include "0324*##" of the NRj. Then, the system increases CF of the translation rule "22732255:8281" and increases EF of the translation rule "4#36569# @a 1:@b 0324*##". Thus, the credibility of "22732255:8281" increases and that of "4#36569# @a 1:@b 0324*##" decreases.

Table 4. Learning Example

NRe	4#36569#227322551
Proofread NRj	828104537##
NR Dictionary	
(English : Japanese)	
4#36569#8366471 : 45304537##	
Acquired NR Translation Rules	
4#36569#8366471 : 45304537##	
4#36569# @a 1 : @b 04537##	
22732255 : 8281	
836647 : 453	
NRj	82810324*##
Re-N-J Translation Process	
Re-N-J Translation Result	*yakiyuuwo* 4 *nasu* 7.
Proofread Translation Result	*yakiyuuwotanosimu.*
Acquired N-J Translation Rules	
8281:*yakiyuu* 0:*wo* 45:*tano*	
37:*simu* ##:.	

Re-N-J Translation Process. The proofread NRj is again translated in order to evaluate only the N-J translation process because the Translation Result in Figure 2 may include errors by the NR translation process. The translation result in this process is expressed as Re-N-J Translation Result in Figure 3.

In Table 4, the proofread NRj is translated into "*yakiyuuwo* 4 *nasu* 7.". "4" and "7" in the Re-N-J translation result have not been translated.

N-J Feedback Process. The N-J dictionary is updated based on the proofread translation result, the proofread NRj and the Re-N-J translation result in this process.

Learning for N-J Dictionary. The translation rules are acquired from the proofread translation result and the proofread NRj, and registered into the N-J dictionary. The proofread translation result is compared to the proofread NRj. When there are common parts between the proofread translation result and the proofread NRj, the common and different parts are registered as new translation rules into the N-J dictionary.

In Table 4, the proofread translation result and the proofread NRj are as follows:

yakiyuuwotanosimu.
828104537##

The underlined segments are common parts because "*wo*" and "*simu*" of the proofread translation result are expressed as *Kana* characters which are Japanese

Table 5. Data of Experiment

Scene	Sentence Number	Character Number for Input	Word Number for Input	Character Number for Proofread
On the Plane	300	7,975	1,562	3,965
At the Airport	550	15,855	2,987	6,927
On the Check-in	400	11,891	2,277	5,366
On the Telephone	250	7,487	1,360	3,328
Total	1,500	43,208	8,186	19,586
Average		28.8	5.5	13.1

phonograms. The segments excluding them are different parts because "*yakiyuu*" and "*tano*" of the proofread translation result are expressed as *Kanji* characters which are Chinese ideographs. The acquired N-J translation rules are shown in Table 4.

Updating for N-J Dictionary. CF and EF for the translation rules are updated based on the Re-N-J translation result and the proofread translation result. If the proofread translation result includes a translation rule of the Re-N-J translation result, the translation rule is correct. Then, the system adds 1 to the value of CF for the translation rule. If the proofread translation result does not include a translation rule of the Re-N-J translation result, the translation rule is erroneous. Then, the system adds 1 to the value of EF for the translation rule. Thus, credibility of the translation rules is updated in the N-J dictionary.

In Table 4, the proofread translation result includes "*yakiyuu*" and "*wo*" of the Re-N-J translation result and does not include "nasu" of the Re-N-J translation result. Then, the system increases CF of the translation rule "8281:*yakiyuu*" and "0:*wo*", and increases EF of the translation rule "53:*nasu*". Thus, the credibility of "8281:*yakiyuu*" and "0:*wo*" increases and that of "53:*nasu*" decreases.

The next translation is performed by the NR dictionary and N-J dictionary which have been updated in the learning process when the user inputs a next English text. Thus, the system based on our proposed method acquires the translation rules and improves because of the repetition of these processes gradually.

4 Evaluation Experiment

The system based on our proposed method has been developed for the experiment. The system translates English into Japanese.

4.1 Data and Procedure

The data for the experiment has four scenes and is shown in Table 5. This data was taken from 10 English travel books for Japanese[7]-[16]. Every English

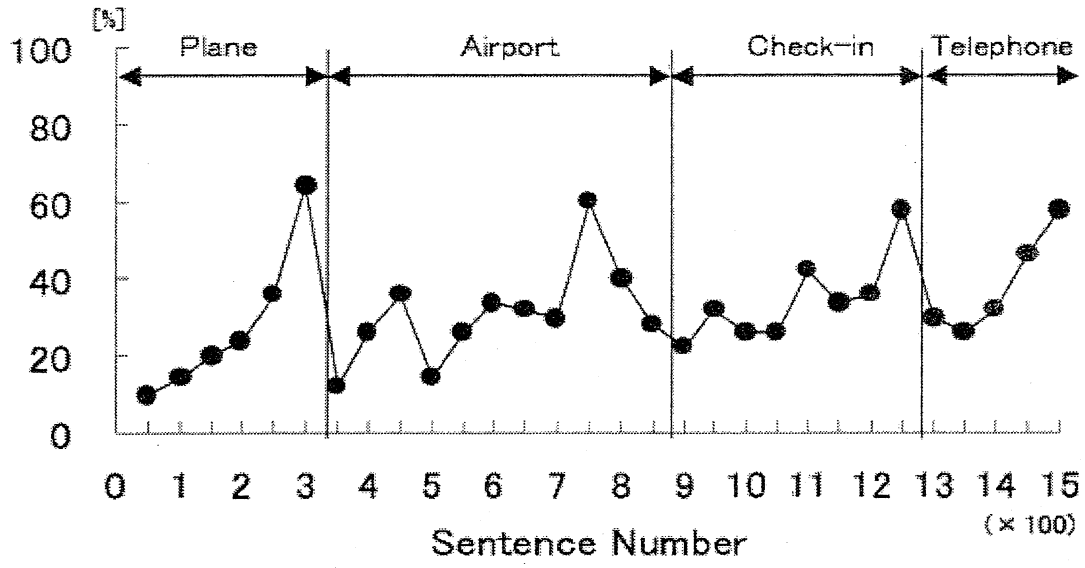


Fig. 4. Change in Correct Translation Rate

text of the data is inputted into the system step by step. If the translation result has errors, they are proofread based on the Japanese text of the data. The initial dictionaries are empty because of evaluation of adaptability of our proposed method. The system is able to improve its own translation performance by Inductive Learning.

We evaluate the correct translation rate per 50 sentences. The correct translation rate is proportion of the number of correct translation sentences to the number of sentences. The translation result was judged by the first author of this paper. The translation result is correct when the meaning of the translation result is identical with that of the Japanese text in the data. In this experiment, the values of the coefficients in equation (1) are $\alpha = 0.1$ and $\beta = 0.5$. It is based on [5][6].

4.2 Results and Considerations

Figure 4 shows the change in the correct translation rate. In Figure 4, the rate of the correct translation increases as the input data increases. When the scene of input data changes, the correct translation rate decreases. The reason is that there are the rules not registered into the NR dictionary and N-J dictionary. However, the correct translation rate increases again because the system acquires the translation rules for the new scene. The last rate of the correct translation is about 60[%] whereas the number of rules in the NR dictionary is only 5,155 and that in the N-J dictionary is only 764. We have also evaluated the translation performance for our proposed method without number representation. The mean of the correct translation rate for this method is lower than that for our proposed method by 4.6[%].

Table 6. Example of NR Rule

NR Dictionary
(English : Japanese) 26853#968# @a #63# @b #8447# @c 1 : 25 @d 40 @e 4144*2732## 8355, 469#86#3455#46, 2273 : 214*5222, 131
English Text Could you tell me how to fill in this card?
NRe 26853#968#8355#63#469#86#3455#46#8447#22731
NRj 25214*522240131414*2732##
Translation Result *konokaadonokakikatawoosieteitadakemasuka?*

The system is able to translate English text into Japanese one which is fit for the scene because of its own adaptability. The translation result for "Beef, please." is "*giyuunikunoriyouriwoonegaisimasu.*" in the scene "on the plane". The translation result is strongly fit for the scene "on the plane".

Since the system acquires NR translation rules which are special for the system, the segmentation of the rules is different from the general one. The translation is performed by the special rules for the system. The example of the rule which is used for the translation is shown in Table 6. In Table 6, the segmentation of the rules in the NR dictionary is different from the general one. However, the NRj is generated by these NR rules. The NRj is able to be translated into the translation result in Table 6. The translation result is correct.

5 Conclusion

In this paper, we evaluated the translation performance in the system based on our proposed method. English is translated into Japanese via number representation on our proposed method. A number representation text is more abstract than the original text because the number representation text corresponds to several texts. Then, the system based on our proposed method is able to acquire more translation rules on number representation than that on the original text by Inductive Learning. Some translation rules include different texts with same meaning. These rules are effective for the translation. The system needs number representation in order to acquire these translation rules. Moreover, the system is able to disambiguate number representation by its own adaptability. Thus, the system translates English into Japanese via number representation. In the evaluation experiment, the last rate of the correct translation is about 60[%] because the system is able to learn the translation rules which are fit for the inputted data and adapt to the present field. The mean of the correct translation rate for our proposed method is higher than that for the method without the number

representation by 4.6[%]. Therefore, it is proved that our proposed method is more effective for machine translation.

A future problem is to tune up the correspondence of number characters with English characters and that with *Kana* characters in order to more improve the translation performance.

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